

Undergraduate Topics in Mathematics 5
Abstract Algebra

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The Fourier Transform

- **The Fourier Transform:** The Fourier transform is a tool that rewrites a function in terms of its frequency content. Instead of describing a signal by “what value it has at each time,” the Fourier transform describes it by “how much of each frequency is present.”

For a function $f(t)$, the FT is written as

$$\hat{f}(\omega) = \mathcal{F}\{f(t)\},$$

and defined by

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t) dt.$$

With inverse

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t} d\omega,$$

which recovers the original function $f(t)$. Note that ω here is the angular frequency.

- **Derivatives become multiplication:** One reason Fourier transforms are so powerful is that they turn differentiation into multiplication.

$$\mathcal{F}\{f'(t)\} = i\omega \hat{f}(\omega).$$

Abstract Algebra

2.1 Relations and functions

- **Relations:** Let S and T be sets. A **relation** from S to T is a subset ρ of $S \times T$, where

$$S \times T = \{(s, t) : s \in S, t \in T\}.$$

If $s \in S$ and $t \in T$, then we write $s\rho t$ if $(s, t) \in \rho$, otherwise we write $s\not\rho t$. In particular, a relation on S is a relation from S to S .

- **Properties of relations:** Let ρ be a relation on S ,
 - **Reflexive:** ρ is reflexive if $a\rho a$ for all $a \in S$
 - **Symmetric:** ρ is symmetric if $a\rho b$ implies $b\rho a$
 - **Transitive:** ρ is transitive if $a\rho b$ and $b\rho c$ implies that $a\rho c$

Note: These properties are defined only for relations from a set to itself.

- **Equivalence relation:** An **equivalence relation** on a set S is a relation that is reflexive, symmetric, and transitive. We use $\sim$ to denote an equivalence relation.

As an example, on $\mathbb{R}$, let us say that $a \sim b$ if and only if $a + b$ is even. First, consider $a, b \in \mathbb{R}$. Observe that

$$a \sim a$$

since whether or not a is even or odd, the sum is always even. So, $\sim$ is reflexive. Next, we notice that if we assume $a \sim b$ (so $a + b$ is even), it immediately follows that $b + a$ is even. So, $b \sim a$ and $\sim$ is symmetric.

Now, let $a, b, c \in \mathbb{R}$. Assume that

$$a + b, b + c$$

are both even. Hence,

$$a + b + b + c = a + 2b + c = a + c + 2b$$

is also even. Notice that $2b$ is even, so we consider two cases, either $a + c$ is odd and $\sim$ is not transitive or $a + c$ is even. If $a + c$ is odd, then we have $a + c = 2\ell + 1$, and

$$a + c + 2b = 2\ell + 1 + 2b = 2(\ell + b) + 1,$$

which is odd. But, we said that $a + c + 2b$ is even, a contradiction. Thus, $a \sim c$ and $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. ■

- **Equivalence class:** Let $\sim$ be an equivalence relation on a set S . If $a \in S$, then the **equivalence class** of a , denoted $[a]$, is the set $\{b \in S : a \sim b\}$
- **Partition:** Let S be a set, and let T be a set of nonempty subsets of S . We say that T is a **partition** of S if every $a \in S$ lies in exactly one set in T .
- **Theorem 1.1:** Let S be a set and $\sim$ be an equivalence relation on S . Then the equivalence classes with respect to $\sim$ form a partition of S . In particular, if $a \in S$, then $a \in [a]$, and $a \in [b]$ if and only if $[a] = [b]$.

Proof. Assume S is a set, and $\sim$ is an equivalence relation on S . To show that the equivalence relations with respect to $\sim$ form a partition of S , we must show that the equivalence relations are a set of nonempty pairwise disjoint subsets of S .

Let $a \in S$. By reflexivity of $\sim$, $a \sim a$. Thus, $a \in [a]$ and the equivalence classes are nonempty. Now, assume that

$$d \in [a] \cap [b].$$

Thus,

$$a \sim d, \quad b \sim d.$$

Now, if $c \in [a]$, then $a \sim c$. Notice that by transitivity,

$$d \sim a, a \sim c \implies c \sim d.$$

Similarly,

$$c \sim d, d \sim b \implies c \sim b.$$

Thus, $c \in [b]$. Since $c \in [a]$ implies that $c \in [b]$,

$$[a] \subseteq [b].$$

A similar argument, except starting with $c \in [b]$ yields that

$$[b] \subseteq [a].$$

Hence, if $d \in [a] \cap [b]$, it follows that $[a] = [b]$.

Therefore, if $a \in S$, then $a \in [a]$, and if $a \in [b]$, then $[b] = [a]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$. ■

- **The partition:** So, now we know that for a equivalence relation $\sim$ on S ,

$$\sim \subseteq S \times S,$$

if $a \in S$, then the equivalence class of a is the set

$$[a] = \{b \in S : a \sim b\}.$$

Further, $b \in [b]$, and so it must be that $[b] = [a]$. In other words, the representative element is not unique for each class.

- **Functions:** Let S and T be any sets. A **function**

$$\alpha : S \rightarrow T$$

is a rule assigning to each $s \in S$ an element $\alpha(s) \in T$.

- **Requirements of a function:** For a relation $f \subseteq A \times B$ to be called a **function** from A to B , it must satisfy

1. **Existence / totality:** For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in f$

$$\forall a \in A, \exists b \in B : (a, b) \in f.$$

2. **Uniqueness / well-definedness:** Each input in A has exactly one output in B

$$\forall a \in A, \forall b_1, b_2 \in B, ((a, b_1) \in f \wedge (a, b_2) \in f) \implies b_1 = b_2.$$

- **Properties of functions:** Consider a function $\alpha : S \rightarrow T$,
 - **Injective:** α is injective if $\alpha(s_1) = \alpha(s_2)$ implies $s_1 = s_2$ for all $s_1, s_2 \in S$
 - **Surjective:** α is surjective if for all $t \in T$, there exists an element $s \in S$ such that $\alpha(s) = t$
 - **Bijective:** α is bijective (invertible) if it is both injective and surjective

We call a bijective function a **bijection**, which is therefore also a **injection** and a **surjection**.

Notice that the inverse of a function only satisfies existence / uniqueness if it is both injective (uniqueness) and surjective (existence).

- **Composition:** Let R, S , and T be sets, and let $\alpha : R \rightarrow S$ and $\beta : S \rightarrow T$ be functions. Then, the **composition** $\beta \circ \alpha$, or simply $\beta\alpha$, is the function from R to T given by

$$(\beta\alpha)(r) = \beta(\alpha(r))$$

for all $r \in R$. Note that $\alpha\beta$ and $\beta\alpha$ are likely different functions.

- **Theorem 1.2: Properties of the composition:** Let $\alpha : R \rightarrow S$, $\beta : S \rightarrow T$, and $\gamma : T \rightarrow U$ be functions. Then,
 1. $(\gamma\beta)\alpha = \gamma(\beta\alpha)$
 2. If α and β are injective, then so is $\beta\alpha$
 3. If α and β are surjective, then so is $\beta\alpha$
 4. If α and β are bijective, then so is $\beta\alpha$

Proof. First, notice that

$$\begin{aligned} (\gamma\beta)\alpha &= \gamma(\beta\alpha), \\ \gamma(\beta\alpha) &= \gamma(\beta\alpha). \end{aligned}$$

Hence, the first property holds. Now, consider $r_1, r_2 \in R$, and assume that

$$\beta(\alpha(r_1)) = \beta(\alpha(r_2)).$$

Since β is injective, $\alpha(r_1) = \alpha(r_2)$, and since α is injective, $r_1 = r_2$. Thus, $\beta\alpha$ is injective.

Next, consider $t \in T$. Since β is surjective, there exists a $s \in S$ such that $\beta(s) = t$. But, α is surjective, so there exists an $r \in R$ such that $\alpha(r) = s$. Thus,

$$(\beta\alpha)r = \beta(\alpha(r)) = \beta(s) = t.$$

Since we showed that $\beta\alpha$ is both injective and surjective, it follows that $\beta\alpha$ is bijective. ■

- **Theorem 1.3: Existence of the inverse:** Let $\alpha : S \rightarrow T$ be a bijection. Then, there exists a bijection $\beta : T \rightarrow S$ such that $(\beta\alpha)(s) = s$ for all $s \in S$ and $(\alpha\beta)(t) = t$ for all $t \in T$.

Proof. Since α is bijective, for all $t \in T$, there exist a unique $s \in S$ such that $\alpha(s) = t$. Now, define $\beta : T \rightarrow S$ such that for that unique s that maps to t ,

$$\beta(t) = s.$$

Then,

$$(\beta\alpha)(s) = \beta(\alpha(s)) = \beta(t) = s.$$

Now, for any $t \in T$, choose $s \in S$ such that $\alpha(s) = t$. By that same property of β ,

$$\beta(t) = s,$$

so

$$(\alpha\beta)(t) = \alpha(\beta(t)) = \alpha(s) = t.$$

Now, we just need to show that β is a bijection. First, assume that $\beta(t_1) = \beta(t_2)$. Then,

$$\begin{aligned}\alpha(\beta(t_1)) &= t_1, \\ \alpha(\beta(t_2)) &= t_2.\end{aligned}$$

But, $\beta(t_1) = \beta(t_2)$, so

$$\alpha(\beta(t_2)) = t_2 = \alpha(\beta(t_1)) = t_1.$$

Thus, β is injective. Next, $s \in S$, we have that

$$\beta(\alpha(s)) = s.$$

Thus, there exists a $t \in T$ such that $\beta(t) = s$ ($t = \alpha(s)$), so β is surjective. ■

- **Permutation:** A permutation of a set S is a bijection from S to S .
- **Binary operation:** Let S be a set. Then, a **binary operation** on S is a function from $S \times S$ to S .

2.2 The Integers and Modular Arithmetic

2.2.1 Induction

- **Property 2.1: Well ordering axiom:** If S is a nonempty set of positive integers, then S has a smallest element.
- **Theorem 2.1: Mathematical induction:** Suppose that for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$

Then, $P(n)$ is true for every positive integer n .

Proof. We proceed by contradiction. Recall that to show a statement $P \implies Q$ is true, it is merely sufficient to show that $P \wedge \neg Q$ is impossible. Thus, if we assume $\neg Q$, and yield a contradiction, $P \wedge \neg Q$ cannot occur and thus the statement is true.

So, we assume that $P(n)$ is not true for every positive integer n . Thus, there is a nonempty set $S \subseteq \mathbb{N}$ such that for every element $n \in S$, $P(n)$ is false. By the well-ordering axiom, there exists a smallest element in S , call this element k .

Observe that the statement assumes that $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$. So, $k > 1$. Further,

$$k-1 \notin S,$$

since if $k-1 \in S$, $k-1+1 = k \notin S$, but k is indeed a member of S . However, since $k-1 \notin S$, $P(k-1)$ is true, and thus $P(k-1+1) = P(k)$ is true, a contradiction.

Therefore, if $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, $P(n+1)$ is also true, then $P(n)$ is true for every positive integer n . ■

Let's consider an example. Suppose we claim that for every positive integer n ,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

First, we observe that

$$1^2 = \frac{1(1+1)(2(1)+1)}{6} = 1.$$

So, $P(1)$ is true. Next, assume that $P(n)$ is true, where

$$P(n) = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Now, observe that

$$\begin{aligned} P(n+1) &= 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 = P(n) + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(2n+1)}{6} + (n+1) \\ &= \frac{n(2n+1) + 6(n+1)}{6} = \frac{2n^2 + n + 6n + 6}{6} = \frac{2n^2 + 7n + 6}{6}. \end{aligned}$$

From here, we can get that

$$\frac{2n^2 + 7n + 6}{6} = \frac{(n+2)(2n+3)}{6}.$$

So,

$$\frac{P(n)}{n+1} + (n+1) = \frac{(n+2)(2n+3)}{6}.$$

Thus,

$$P(n) + (n+1)^2 = P(n+1) = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}$$

Thus, $P(n)$ implies $P(n+1)$, and P therefore holds for every positive integer n .

- **Binomial theorem:** Let a and b be real numbers, and n a positive integer. Then,

$$\begin{aligned} (a+b)^n &= a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + b^n \\ &= \sum_{k=0}^{n-1} \binom{n}{k} a^{n-k} b^k. \end{aligned}$$

Proof. *prove by induction on n .*

- **Strong induction:** Suppose that, for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each integer $n > 1$, if $P(k)$ is true for every $k < n$, then $P(n)$ is true.

Then, $P(n)$ is true for every positive integer n .

Proof. Assume that the two conditions hold, but $P(n)$ is not true for every positive integer n . Then, let S be the set of integers in which $P(s)$ is false. Since $S \subseteq \mathbb{N}$ and is nonempty, here is a smallest element j by the well-ordering axiom. Call this smallest element ℓ .

Note that since $P(1)$ is true, $\ell > 1$. Since ℓ is the smallest element in S , and therefore the first element that makes the proposition false, it must be that $P(k)$ is true for all $k < \ell$. But, that would imply that $P(\ell)$ is in fact true, a contradiction.

Thus, there is no such set of elements S that make P false, so $P(n)$ true for all $n \in \mathbb{N}$.

2.2.2 Some Number Theory

- **Theorem 2.4 (Division algorithm):** Let $a, b \in \mathbb{Z}$ with $b > 0$. Then, there exists unique integers q and r such that

$$a = bq + r, \quad 0 \leq r < b.$$

We call q and r the **quotient** and **remainder** respectively.

Proof. First, let

$$S = \{a - bt : t \in \mathbb{Z}, a - bt \geq 0\}.$$

First, if $0 \in S$, then $a - bq = 0$ for some $q \in \mathbb{Z}$, and so

$$a = bq + 0$$

and the theorem holds. So, $S \subseteq \mathbb{N}$. We first claim that S is nonempty. Let

$$t = -|a|.$$

Then,

$$a - bt = a + |a|b.$$

If $a \geq 0$, then $a + |a|b \geq 0$ since $b > 0$. If $a \leq 0$ then

$$a + |a|b = a - ab = a(1 - b).$$

Notice that since $b > 0, b \in \mathbb{Z}$, it must be that $b \geq 1$. Thus,

$$1 - b \leq 0.$$

So, $a(1 - b) \geq 0$. Thus, we have shown the existence of at least one t such that $a - bt \in S$, and so S is nonempty.

Since S is a nonempty subset of the natural numbers, we have a least element by the well-ordering axiom, say

$$r = a - bq.$$

Recall that $r \geq 0$ by definition. Suppose that $r \geq b$, then $0 \leq r - b < r$. But,

$$r - b = a - bq + b = a - b(q + 1) < r,$$

a contradiction, since r is the smallest element. Note that

$$r - b = a - b(q + 1)$$

is indeed a member of S , since $r - b \geq 0$. Thus,

$$a = bq + r, \quad 0 \leq r < b.$$

For uniqueness, suppose that

$$a = bq_1 + r_1 = bq_2 + r_2$$

with $q_i, r_i \in \mathbb{Z}$ and $0 \leq r_i < b$. Then,

$$b(q_1 + q_2) = r_1 - r_2.$$

If we take the absolute value,

$$|b(q_1 + q_2)| = |r_1 - r_2|.$$

But, $b > 0$, so

$$b|q_1 - q_2| = |r_2 - r_1|.$$

Notice that since $q_1, q_2 \in \mathbb{Z}$, $q_1 - q_2 \in \mathbb{Z}$, so if $q_1 \neq q_2$,

$$b|q_1 - q_2| \geq b.$$

But, $0 \leq r_1, r_2 < b$, so

$$|r_2 - r_1| < b,$$

and $|r_2 - r_1| = b|q_1 - q_2|$, which we just said was $\geq b$, a contradiction. Hence, $q_1 = q_2$. In that case,

$$b(q_1 - q_2) = b(0) = r_2 - r_1.$$

Thus, $r_1 = r_2$. ■.

- **Divides:** Let a and b be integers, we say that a divides b if there exists an integer c such that $b = ac$, in which case we write

$$a \mid b.$$

If no such c exists, we write $a \nmid b$

- **Lemma 2.1:** Let $a, b, c \in \mathbb{Z}$. Then,
 1. if $a \mid b$ and $b \mid c$, then $a \mid c$
 2. if $a \mid b$ and $b \neq 0$, then $a \leq |b|$
 3. if $a \mid b$ and $a \mid c$, then $a \mid (bu + cv)$ for any $u, v \in \mathbb{Z}$

Proof. First, assume that $a, b, c \in \mathbb{Z}$. (1). Assume that $a \mid b$, and $b \mid c$. Then,

$$b = a\ell, \quad c = bk.$$

If $k = 0$, then $c = 0$, and $a \mid c$. So, assume that $k \neq 0$. Then,

$$b = \frac{c}{k},$$

and so

$$\frac{c}{k} = a\ell.$$

Hence, $c = a(\ell k)$, and $a \mid c$.

(2) Assume that $a \mid b$, $b \neq 0$. So,

$$b = ar, \quad r \in \mathbb{Z}.$$

Consider the case that $a \leq |b|$ is false, so $|b| < a$, or

$$-a < b < a.$$

Notice that $a = 0$ would imply that $b = 0$, but $b \neq 0$. So, $a \neq 0$. Then,

$$-1 < \frac{b}{a} = r < 1.$$

But, this would contradict the fact that $r \in \mathbb{Z}$. So, $a \leq |b|$.

(3). Suppose that $a \mid b$ and $a \mid c$, so

$$b = ar, \quad c = ak, \quad r, k \in \mathbb{Z}.$$

Let $u, v \in \mathbb{Z}$,

$$bu = aru, \quad cv = akv.$$

Then,

$$bu + cv = aru + akv = a(ru + kv).$$

Since $ru + kv \in \mathbb{Z}$, $a \mid bu + cv$ for all $u, v \in \mathbb{Z}$. ■

- **Greatest common divisor:** Let a and b be integers, not both zero. Then, the **greatest common divisor (gcd)** of a and b , written (a, b) , is the largest positive integer g such that $g \mid a$ and $g \mid b$.

Notice that (a, b) is always at least 1, and $(a, b) \leq |a|$. So, only numbers from 1 to $|a|$ are considered.

- **Lemma 2.2:** Take any integers a, b with $a \neq 0$. Then,

1. $(a, b) = (-a, b)$
2. $(a, 0) = |a|$

Proof. Assume a, b are integers with $a \neq 0$.

(1). Assume that $(a, b) = g$, so

$$a = gk, \quad k \in \mathbb{Z}.$$

In that case,

$$-a = g(-k).$$

Since $-k \in \mathbb{Z}$, $g \mid -a$. Trivially, g is still maximal for $(-a, b)$.

(2). Since all numbers divide 0, and $d \leq |a|$ for any number d that divides a , the GCD is therefore $|a|$. ■

- **Relatively prime:** Let $a, b \in \mathbb{Z}$, not both zero. Then, we say that a, b are **relatively prime** if $(a, b) = 1$.
- **Theorem 2.5:** Let a and b be integers, not both 0. Then, there exists integers u, v such that

$$(a, b) = au + bv.$$

Furthermore, (a, b) is the smallest positive integer that can be written in this way.

Proof. Let

$$S = \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\} \subseteq \mathbb{N}.$$

First, we show that S has at least one element. If we assume $a \neq 0$, and let $x = a$, $b = 0$, then

$$a(a) + b(0) = a^2 > 0.$$

So, $S \neq \emptyset$. By the well-ordering axiom, S has a least element, say

$$g = au + bv.$$

If we show that $g = (a, b)$, the proof is complete.

Suppose that a and b has some only common divisor c , then

$$c \mid a \wedge c \mid b.$$

So,

$$c \mid au + bv$$

for all $u, v \in \mathbb{Z}$. Hence, $c \mid g$, and $c \leq g$. That is, g would indeed be the greatest common divisor provided we can show that $g \mid a$ and $g \mid b$.

Using the division algorithm,

$$a = gq + r, \quad 0 \leq r < g.$$

So,

$$r = a - gq = a - (au + bv)q = a - auq - bvq = a(1 - uq) + b(-vq).$$

Hence, $r \in S$ if $r > 0$. But, $r < g$, contradicting the minimality of g . So, $r = 0$, and $g \mid a$. Using the same argument, we can also get that $g \mid b$. ■

- **Corollary 2.1:** Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. Then, a and b are relatively prime if and only if there exists integers u, v such that

$$au + bv = 1.$$

- **Corollary 2.2:** Let $a, b, c \in \mathbb{Z}$ with $\{a, b\} \neq \{0\}$. If $(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Proof. We have

$$au + bv = 1, \quad u, v \in \mathbb{Z}.$$

So,

$$auc + bvc = c.$$

Since $a \mid a$ and $a \mid bc$,

$$a \mid a(uc) + bc(v),$$

since $uc, v \in \mathbb{Z}$. Thus, $a \mid c$. ■

- **Corollary 2.3:** Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. If $(a, b) = 1$, and for same integer n we have $a \mid n$ and $b \mid n$, then $ab \mid n$.
- **Theorem 2.6: Euclidean algorithm:** Let a, b be integers, with b positive. If $b \mid a$, then $(a, b) = b$. Otherwise, apply the division algorithm repeatedly. Let

$$\begin{aligned} a &= bq_1 + r \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \\ r_{k-2} &= r_{k-1}q_k + r_k \\ r_{k-1} &= r_kq_{k+1} + 0, \end{aligned}$$

where $q_i, r_i \in \mathbb{Z}$ for all i , and $0 < r_k < r_{k-1} < \dots < r_1 < b$. Then, $(a, b) = r_k$.

Note: You may have seen this algorithm defined recursively as

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Proof. If $b \mid a$, then $b \leq |a|$. Note that $b \mid b$, so b is trivially the greatest divisor of b . Thus, since $b \leq a$ and b is the greatest divisor of b , it should be clear that $(a, b) = b$. Thus, we may assume that $b \nmid a$.

By the division algorithm,

$$a = bq_1 + r_1, \quad q_1, r_1 \in \mathbb{Z}.$$

Notice that $r_1 = a - bq_1$. Assume $c \mid a$ and $c \mid b$. Then,

$$c \mid a(1) + b(-q_1) = a - bq_1 = r_1.$$

Thus, any common divisor of a and b is also a common divisor of b and r_1 , with $r_1 = a \bmod b$. Conversely, if $d \mid b$ and $d \mid r_1$, then

$$d \mid b(q_1) + r_1(1) = a.$$

So, any common divisor of b and r_1 is also a common divisor of a and b . Hence, $b, a \bmod b$ have precisely the same common divisors as a and b .

Also, observe that

$$0 < r_1 < b.$$

If $r_1 = 0$, then $b \mid a$, so $r_1 > 0$. Now, we can repeat this argument to say that the common divisors of b, r_1 are the same as the common divisors of r_1, r_2 , where $r_2 = b \bmod r_1 = b \bmod a \bmod b$, with

$$0 < r_2 < r_1.$$

Thus,

$$(a, b) = (b, r_1) = (r_1, r_2) = \cdots = (r_{k-1}, r_k) = (r_k, 0) = r_k.$$

Since $0 < r_k < r_{k-1} < \cdots < r_2 < r_1 < b$, each iteration produces a smaller r_i than the one prior, and since all are greater than zero, the sequence must terminate in finite steps.

Note: We note that if $b < 0$, we cannot use the division algorithm, as that requires $b > 0$. This is why we assume b is positive for this theorem. However, recall that

$$(a, b) = (-a, b).$$

This fact, along with $(a, b) = (b, a)$ lets us use the algorithm with

$$(a, -b)$$

to find the gcd of (a, b) .

- **Using the Euclidean algorithm:** Note that since

$$r_{k-2} = r_{k-1}q_k + r_k \implies (a, b) = r_k = r_{k-2} - r_{k-1}q_k = r_{k-2}(1) + r_{k-1}(-q_k),$$

with

$$\begin{aligned}
a &= bq_1 + r_1, \\
b &= r_1q_2 + r_2, \\
r_1 &= r_2q_3 + r_3, \\
&\vdots \\
r_{k-4} &= r_{k-3}q_{k-2} + r_{k-2}, \\
r_{k-3} &= r_{k-2}q_{k-1} + r_{k-1},
\end{aligned}$$

we can work our way backwards to get integers u, v such that

$$(a, b) = au + bv.$$

For example, consider $a = 45$, $b = 33$. Applying the Euclidean algorithm yields

$$\begin{aligned}
45 &= 33(1) + 12, \\
33 &= 12(2) + 9, \\
12 &= 9(1) + 3, \\
9 &= 3(3) + 0.
\end{aligned}$$

Hence, $(a, b) = 3$. Now, we can find u, v such that $au + bv = 3$. We have

$$\begin{aligned}
3 &= r_1(1) + r_2(-q_3) = 12(1) + 9(-1), \\
&= r_1(1) + (b - r_1q_2)(-q_3) = 12(1) + (33 - 12(-2))(-1) \\
&= (a - bq_1) + b(-q_3) + r_1q_2q_3 \\
&= (a - bq_1) + b(-q_3) + (a - bq_1)q_2q_3 \\
&= a + b(-q_1) + b(-q_3) + aq_2q_3 + b(-q_1)q_2q_3 \\
&= a(1 + q_2q_3) + b(-q_1 - q_3 - q_1q_2q_3) \\
&= 45(1 + 2(1)) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-4).
\end{aligned}$$

Hence, $u = 3$, and $v = -4$.

- **Prime numbers:** A natural number $p > 1$ is said to be prime if its only positive divisors are 1 and p . Otherwise, it is composite. Note that 1 is neither prime nor composite.
- **Euclid's Lemma:** Let $p > 1$ be a positive integer. Then, the following are equivalent.
 1. p is prime; and
 2. if a and b are integers such that $p \mid ab$, then $p \mid a$ or $p \mid b$

Proof. First, suppose that p is prime and $p \mid ab$. Either $(p, a) = 1$ or $(p, a) = p$. If $p \leq a$, we have $(p, a) = p$, and so p is a common divisor of both p and a , hence $p \mid a$.

If $p > a$, then $(p, a) = 1$. In that case,

$$pu + av = 1,$$

so

$$pub + avb = b.$$

Now, with $p \mid p$ and $p \mid ab$,

$$p \mid pk + abc, \quad k, c \in \mathbb{Z}.$$

If we let $k = ub$, $c = v$, then

$$p \mid p(ub) + abv = b.$$

So, $p \mid b$. Now, consider the case in which p is composite. In that case, let $p = cd$. Since $p > 1$,

$$1 < c, d < p.$$

So, $p \nmid c$ and $p \nmid d$. ■

- **Corollary 2.4:** Let p be a prime number and $a_1, \dots, a_n \in \mathbb{Z}$. If

$$p \mid a_1 a_2 \cdots a_n,$$

then $p \mid a_i$ for some i . In fact, every positive integer larger than 1 can be written as a product of primes, called its **prime factorization**.

Proof. Assume $p \in \mathbb{P}$, and $a_1, a_2, \dots, a_n \in \mathbb{Z}$. Further assume that $p \mid a_1 a_2 \cdots a_n$.

So, $p \mid a_1$ or $p \mid a_2 \cdots a_n$. Further, $p \mid a_2$ or $p \mid a_3 \cdots a_n$. We can extend this idea up to $p \mid a_{n-1}$ or $p \mid a_n$. Thus, $p \mid a_i$ for some i . ■

- **Theorem 2.8: Fundamental Theorem of Arithmetic:** If $a \in \mathbb{N}$ and $a > 1$, then there exists primes $p_1, \dots, p_n$ not necessarily distinct such that $a = p_1 p_2 \cdots p_n$. Further, this product is unique up to order. That is, if $a = q_1 q_2 \cdots q_m$ for some primes q_i , then $m = n$, and after rearranging the primes, $p_i = q_i$ for all i .

Proof. Notice that we start with $a = 2$, which is prime. Thus, let $a > 2$ and assume that the theorem is true for smaller numbers. If a is prime, nothing to do. If a is composite, then we can write $a = bc$ for $1 < b, c < a$. Since $b, c < a$, b and c can be written as a product of primes. Thus, a is a product of primes.

Now we prove uniqueness. Suppose that

$$a = p_1 \cdots p_n = q_1 \cdots q_m$$

for some primes p_i and q_i . Without loss of generality, say $n \leq m$. Since

$$p_1 \mid a,$$

$p_1 \mid q_i$ for some i . We can rearrange the order to say that $p_1 \mid q_1$. But, q_1 is prime, so $p_1 = 1$ or $p_1 = q_1$. Since 1 is not prime, $p_1 = q_1$. So, we can cancel p_1 from both sides to get

$$p_2 \cdots p_n = q_2 \cdots q_m.$$

We can continue this process, finding that $p_i = q_i$ for $1 \leq i \leq n$. If $m = n$, we are done. Otherwise, we are left with $1 = q_{n+1} \cdots q_m$. But then $q_m \mid 1$, which is impossible since $q_m > 1$. ■

- **Corollary 2.5:** Let a, b , and n be integers, with $n \neq 0$. If $(a, n) = (b, n) = 1$, then $(ab, n) = 1$.

Proof. Assume that $a, b, n \in \mathbb{Z}$, $n \neq 0$, $(a, n) = (b, n) = 1$.

Assume that $g = (ab, n) > 1$. By the Fundamental Theorem of Arithmetic, g can be factorized as a product of primes. Hence, there exists a prime p that divides g . Since g divides ab and p divides g , $p \mid ab$. In that case, $p \mid a$ or $p \mid b$. But, $p \mid n$ by similar logic. So,

$$(a, n) \geq p \quad \text{or} \quad (b, n) \geq p.$$

Either way, we have a contradiction, since $(a, n) = (b, n) = 1$. ■.

2.2.3 Properties of the integers

- **Intro:** We note that our discussion will take place in $\mathbb{Z}$, but it could just as easily use $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

- **Closure:** $\mathbb{Z}$ is closed under addition and multiplication

$$a + b, ab \in \mathbb{Z}$$

for all $a, b \in \mathbb{Z}$.

- **Associativity:** Addition and multiplication on $\mathbb{Z}$ are both associative.

$$\begin{aligned}(a + b) + c &= a + (b + c), \\ (ab)c &= a(bc)\end{aligned}$$

for all $a, b, c \in \mathbb{Z}$.

- **Commutative:** Addition and multiplication are both commutative on $\mathbb{Z}$.

$$\begin{aligned}a + b &= b + a, \\ ab &= ba\end{aligned}$$

for all $a, b \in \mathbb{Z}$.

- **Identity:** We call 0 the additive identity for $\mathbb{Z}$, and 1 the multiplicative identity for $\mathbb{Z}$

$$a + 0 = a, \quad 1a = a$$

for all $a \in \mathbb{Z}$.

- **Inverse:** If $a \in \mathbb{Z}$, then $-a$ is its additive inverse.

$$a + (-a) = 0.$$

We note that there is no multiplicative inverse in $\mathbb{Z}$, unlike $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

2.2.4 Modular arithmetic

- **Intro:** Consider $a, b, n \in \mathbb{Z}$. By the division algorithm,

$$\begin{aligned} a &= nc + r_1, \\ b &= nk + r_2 \end{aligned}$$

for $c, k, r_1, r_2 \in \mathbb{Z}$, $0 \leq r_1, r_2 < n$. Now, let's consider $a - b$,

$$a - b = nc + r_1 - nk - r_2 = n(c - k) + (r_1 - r_2).$$

Notice that if $n \mid a - b$, then $r_1 - r_2 = 0$, so $r_1 = r_2$. Thus, if $n \mid a - b$, a and b have the same remainder when divided by n .

- **Congruence:** Let $n \geq 2$ be an integer. If $a, b \in \mathbb{Z}$, then we say that a is congruent to b modulo n , and write $a \equiv b \pmod{n}$ provided

$$n \mid a - b.$$

That is, a and b have the same remainder when divided by n .

- **Reduction to the remainder:** Suppose that $a, b, n \in \mathbb{Z}$, $n \geq 2$, and $n \mid a - b$. So,

$$a \equiv b \pmod{n}.$$

In that case, a and b have the same remainder when divided by n , so

$$\begin{aligned} a &= nc + r, \\ b &= nk + r \end{aligned}$$

for $c, k \in \mathbb{Z}$. Notice

$$\begin{aligned} a - r &= nc, \\ b - r &= nk. \end{aligned}$$

So, $n \mid a - r$, and $n \mid b - r$. Thus, we can reduce to

$$\begin{aligned} a &\equiv r \pmod{n}, \\ b &\equiv r \pmod{n}. \end{aligned}$$

- **Theorem 2.9:** Let $n \geq 2$ be an integer. Then, $a \equiv b \pmod{n}$ is an equivalence relation on $\mathbb{Z}$. The equivalence class of a consists of all integers having the same remainder as a when divided by n .

Proof. Assume $n \geq 2$ is an integer, let $a \equiv b \pmod{n}$ be a relation on $\mathbb{Z}$. First, observe that

$$n \mid 0 = a - a.$$

Thus, $a \equiv a \pmod{n}$, and the relation is reflexive. Next, Assume that

$$a \equiv b \pmod{n}.$$

Thus, $n \mid a - b$, or

$$a - b = nc$$

for some $c \in \mathbb{Z}$. Then,

$$-(b - a) = nc \implies b - a = n(-c).$$

Thus, $n \mid b - a$, so $b \equiv a \pmod{n}$ and the relation is symmetric. Last, assume that

$$a \equiv b \pmod{n}, \quad b \equiv c \pmod{n}.$$

Then,

$$a - b = nr, \quad b - c = nk$$

for some $r, k \in \mathbb{Z}$. Thus,

$$a - b + b - c = a - c = nr + nk = n(r + k).$$

Thus, $n \mid a - c$, so $a \equiv c \pmod{n}$ and the relation is transitive. Hence, we have an equivalence relation on $\mathbb{Z}$.

And, as we know for an equivalence relation $\sim \subseteq S \times S$,

$$[a] = \{b \in S : a \sim b\},$$

and if $b \in [a]$, $[a] = [b]$. Thus, for the equivalence relation $a \equiv b \pmod{n}$, let $a \in \mathbb{Z}$, then the equivalence class $[a]$ is

$$[a] = \{b \in \mathbb{Z} : a \equiv b \pmod{n}\}.$$

So, the integers $b \in \mathbb{Z}$ with the same remainder when divided by n makes up the equivalence class for a . ■

- **Definition 2.6:** Let $n \geq 2$ be an integer. The set of integers modulo n , denoted $\mathbb{Z}_n$ is the set of equivalence classes of $\mathbb{Z}$ with respect to the equivalence relation $a \equiv b \pmod{n}$. We call these the **congruence classes** modulo n . Specifically,

$$\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}.$$

Recall that the representative of a class is not unique. However, it is customary to reduce final answers in $\mathbb{Z}_n$ to the form $[a]$, where $0 \leq a < n$.

- **Residue class:** The term **residue class** is another name for **congruence class**. A **residue** is usually a representative of that class.
- **Sumsets:** Consider $S, T \subseteq \mathbb{Z}$. Then,

$$S + T = \{s + t : s \in S, t \in T\}$$

is called the **sumset**. Similarly, we can define the **difference set**, **product set**, or **quotient set** in a similar way, just changing the operation.

- **Addition and multiplication on $\mathbb{Z}_n$:** We have

$$\begin{aligned} [a] + [b] &= [a + b], \\ [a][b] &= [ab]. \end{aligned}$$

Consider the sumset

$$[a] + [b] = \{x + y : x \in [a], y \in [b]\}.$$

Note that

$$[a + b] = \{s \in \mathbb{Z} : a + b \equiv s \pmod{n}\}.$$

So, we are asserting that

$$\{x + y : x \in [a], y \in [b]\} = \{s \in \mathbb{Z} : a + b \equiv s \pmod{n}\}.$$

Proof. Let $x + y = [a] + [b]$, so $x \in [a]$, and $y \in [b]$. Hence,

$$\begin{aligned} a &\equiv x \pmod{n}, \\ b &\equiv y \pmod{n}. \end{aligned}$$

Thus,

$$a - x = nc, \quad b - y = nk$$

for some $c, k \in \mathbb{Z}$. Thus,

$$a - x + b - y = a + b - (x + y) = n(c + k).$$

Hence,

$$a + b \equiv x + y \pmod{n}.$$

So, $x + y \in [a + b]$, and $[a] + [b] \subseteq [a + b]$. Now, let $z \in [a + b]$, so

$$z \equiv a + b \pmod{n}.$$

Let $x = a$, and $y = z - a$. Then, $x + y = z$,

$$x \equiv a \pmod{n},$$

and

$$z - a = y \equiv (a + b) - a \pmod{n}.$$

So, $y \equiv b \pmod{n}$, hence $x \in [a]$ and $y \in [b]$. Therefore $[a + b] \subseteq [a] + [b]$, and we can conclude that

$$[a] + [b] = [a + b].$$

■

Now, consider

$$\begin{aligned} [a][b] &= \{xy : x \in [a], y \in [b]\}, \\ [a \cdot b] &= \{z \in \mathbb{Z} : ab \equiv z \pmod{n}\}. \end{aligned}$$

We assert that

$$\{xy : x \in [a], y \in [b]\} = \{z \in \mathbb{Z} : ab \equiv z \pmod{n}\}.$$

The proof for this is non-trivial. Instead of proving these two operations, we can instead define them as such, and show that they are well-defined.

- **A well-defined operation on a set:** An operation is well-defined on a set if it gives a single, unambiguous output in that set for every allowed input from that set. For a binary operation $*$ on a set S ,

$$* : S \times S \rightarrow S.$$

$*$ is well-defined on S if for every $a, b \in S$, $a * b$ exists, is unique, and is also an element of S .

- **Theorem 2.10:** For any integer $n \geq 2$, addition and multiplication on $\mathbb{Z}_n$ are well-defined.

Proof. Suppose that $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_2 \pmod{n}$. Then,

$$\begin{aligned} a_1 - a_2 &= nc, \\ b_1 - b_2 &= nk. \end{aligned}$$

So,

$$a_1 + b_1 - (a_2 + b_2) = n(c + k).$$

Thus,

$$a_1 + b_1 \equiv a_2 + b_2 \pmod{n}.$$

Hence, $[a_1 + b_1] = [a_2 + b_2]$, and we can say that addition is well-defined.

Now,

$$\begin{aligned} a_1 &= nc + a_2, \\ b_1 &= nk + b_2. \end{aligned}$$

So,

$$a_1 b_1 = (nc + a_2)(nk + b_2) = n^2 ck + ncb_2 + nka_2 + a_2 b_2.$$

Hence,

$$a_1 b_1 - a_2 b_2 = n(nck + cb_2 + ka_2),$$

and we can say that $a_1 b_1 \equiv a_2 b_2$, so

$$[a_1 b_1] = [a_2 b_2].$$

Therefore, multiplication on residue classes is also well-defined.

- **Theorem 2.11:** Let $n \geq 2$ be an integer, and take any $[a], [b], [c] \in \mathbb{Z}_n$. Then,

1. $[a] + [b] \in \mathbb{Z}_n$
2. $[a] + ([b] + [c]) = ([a] + [b]) + [c]$
3. $[a] + [b] = [b] + [a]$
4. $[a] + [0] = [a]$
5. $[a] + [-a] = [0]$

- **Theorem 2.12:** Let $n \geq 2$ be an integer, and $[a], [b], [c] \in \mathbb{Z}_n$, then

1. $[a][b] \in \mathbb{Z}_n$
2. $[a]([b][c]) = ([a][b])[c]$
3. $[a][b] = [b][a]$
4. $[a]([b] + [c]) = [a][b] + [a][c]$
5. $[1][a] = [a]$

- **Notational switch:** When working in $\mathbb{Z}_n$, we will normally simply write a or $a + b$ instead of $[a]$ or $[a] + [b]$, as long as the context is clear.

2.3 Groups

2.3.1 Introduction to groups

- **Permutations:** Let A be the set $\{1, 2, 3\}$. We would like to consider all the permutations of A . The notation we will use is

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ a & b & c \end{pmatrix},$$

which means $\sigma(1) = a, \sigma(2) = b$, and $\sigma(3) = c$.

- **Composition of permutations:** Consider

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Then,

$$\begin{aligned} (\sigma \circ \tau)(1) &= \sigma(\tau(1)) = \sigma(3) = 3, \\ (\sigma \circ \tau)(2) &= \sigma(\tau(2)) = \sigma(1) = 2, \\ (\sigma \circ \tau)(3) &= \sigma(\tau(3)) = \sigma(2) = 1. \end{aligned}$$

Thus,

$$\sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

- **Properties of permutations:**

1. **Closure:** Permutations are closed under composition
2. **Associativity:** For permutations σ, τ, ρ , we have $\rho \circ (\sigma \circ \tau) = (\rho \circ \sigma) \circ \tau$
3. **Identity:** If σ is any permutation of A , then

$$\sigma \circ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \circ \sigma = \sigma.$$

4. **Inverse:** For each permutation σ , there is another permutation τ such that

$$\sigma \circ \tau = \tau \circ \sigma = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Note that we do not have commutativity, $\sigma \circ \tau \neq \tau \circ \sigma$ in general.

- **Symmetric group on n letters, S_n :** Permutations under composition give us an example of a group. If we let $A = \{1, 2, 3, \dots, n\}$, for any positive integer n , then we call the set of all permutations of this set, under the composition operation the **symmetric group on n letters**, denoted S_n .
- **Groups:** A **group** is a set G , together with a binary operation $*$, satisfying the following conditions
 1. $a * b \in G$ for all $a, b \in G$ (closure);
 2. $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$ (associativity);
 3. there exists an $e \in G$ such that $a * e = e * a = a$ for all $a \in G$ (existence of identity); and
 4. for each $a \in G$, there exists a $b \in G$ such that $a * b = b * a = e$ (existence of inverses).

We refer to G as a group **under** $*$. The element e is called the **identity** of the group. Also, if b is the inverse of a , then we write $b = a^{-1}$.

- **Abelian group:** A group G is said to be **abelian** if $a * b = b * a$ for all $a, b \in G$. So, $*$ is commutative.

Note that $\mathbb{Z}$ and $\mathbb{Z}_n$ for $n \geq 2$ are abelian groups under addition. In fact, 0 is the identity, and the inverse of a is $-a$. The same can be said for $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ under addition.

- **Group tables:** When a group G has only finitely many elements, we can represent it with a **group table**. We write the elements of G down the first column and along the first row of the table. Then, the entry in the row headed by a and the column headed by b is $a * b$. Observe the group table for the additive group $\mathbb{Z}_5$

	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

- **Example:** Let G be the set of nonzero rational numbers. Then, G is an abelian group under multiplication.

Proof. Let

$$G = \{q \in \mathbb{Q} : q \neq 0\} \subseteq \mathbb{Q}.$$

Consider $q_1, q_2, q_3 \in G$, with

$$q_1 = \frac{a}{b}, q_2 = \frac{c}{d}, q_3 = \frac{e}{f}.$$

Then,

$$q_1 q_2 = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

It should be clear that $a, b, c, d \neq 0$, so $q_1 q_2 \in G$. Thus, we have closure. Next,

$$q_1 \cdot (q_2 \cdot q_3) = (q_1 \cdot q_2) \cdot q_3$$

since multiplication is associative. Also,

$$q_1 q_2 = q_2 q_1$$

since multiplication is commutative. Clearly $e = 1$ is the identity, since $1 \in G$ and $1q = q$ for all $q \in G$. Also, for

$$q = \frac{a}{b} \in G,$$

$$q \cdot \frac{1}{q} = \frac{a}{b} \cdot \frac{b}{a} = 1.$$

Thus, $q^{-1} = \frac{1}{q}$.

- **Example:** Let $n \geq 2$ be positive integer. Let $U(n)$ denote the set of all elements $a \in \mathbb{Z}_n$ such that $(a, n) = 1$.